Speed Control of a BDC Motor with Vertical Rod: From First-Principles Dynamics to Data-Driven Feedforward

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1. Motor Dynamics

A brushed DC (BDC) motor driving a vertical rigid rod is modelled through two coupled subsystems.

Electrical subsystem

Applying Kirchhoff's voltage law to the armature circuit:

$$L\,\frac{di}{dt} = V_{\text{in}} - Ri - K_b\,\omega \tag{1}$$

where V_{in} is the applied voltage, i the armature current, ω the shaft angular speed, R the armature resistance, L the inductance, and K_b the back-EMF constant.

Mechanical subsystem

The net torque balance on the motor shaft with an attached rod of mass m, length ℓ, centre of mass at $\ell_{cm} = \ell/2$, and moment of inertia $I_{\text{rod}} = \frac{1}{3}m\ell^2$ gives:

$$J_{\text{eff}}\,\frac{d\omega}{dt} = K_t\,i - B\,\omega - mg\ell_{cm}\sin\theta \tag{2}$$

where $J_{\text{eff}} = J_m + I_{\text{rod}}$, K_t the torque constant, B the viscous friction coefficient, θ the rod angle from the downward vertical, and g gravitational acceleration. The system is integrated numerically using a fixed-step RK4 integrator with $\Delta t = 10^{-3}$ s.

The parameters used throughout are listed in Table 1.

Table 1: Simulation parameters

Symbol	Value	Symbol	Value
R	$3.0\,\Omega$	J_m	$7.4\times10^{-5}\,\mathrm{kg\,m^2}$
L	$4\,\mathrm{mH}$	B	$5\times10^{-3}\,\mathrm{N\,m\,s/rad}$
K_t	$0.05\,\mathrm{N\,m/A}$	m	$0.05\,\mathrm{kg}$
K_b	$0.05\,\mathrm{V\,s/rad}$	ℓ	$0.1\,\mathrm{m}$
$V_{\max}$	$12\,\mathrm{V}$	g	$9.81\,\mathrm{m/s^2}$

2. Gravity Compensation at Steady State

To derive a feedforward term for gravity, consider the steady-state condition where all derivatives vanish ($\frac{di}{dt} = 0,\ \frac{d\omega}{dt} = 0$):

$$V = Ri_{ss} + K_b\,\omega \tag{3}$$

$$0 = K_t\,i_{ss} - B\,\omega - mg\ell_{cm}\sin\theta \tag{4}$$

Solving (4) for i_{ss} and substituting into (3):

$$V_{ss} = \underbrace{\frac{R\,mg\ell_{cm}\sin\theta}{K_t}}_{V_{\text{grav}}} + \underbrace{\left(\frac{RB}{K_t} + K_b\right)\omega}_{V_\omega} \tag{5}$$

The gravity feedforward term V_{grav} is extracted and added directly to the PI control output:

$$V_{\text{cmd}} = V_{\text{grav}} + K_p\,e + K_i\int e\,dt \tag{6}$$

3. Inverse Model via Laplace Transform

Taking the Laplace transform of (1) and (2) and eliminating $I(s)$:

$$V_{\text{in}}(s) = \underbrace{\frac{(J_{\text{eff}}s + B)(Ls + R) + K_tK_b}{K_t}}_{G_1(s)}\,\Omega(s) + \underbrace{\frac{Ls + R}{K_t}}_{G_2(s)}\,T_{\text{load}}(s) \tag{7}$$

where $T_{\text{load}}(s) = mg\ell_{cm}\sin\theta$ is treated as a disturbance. The term $G_1(s)$ gives the voltage required to produce a desired speed trajectory $\Omega_{\text{ref}}(s)$, while $G_2(s)$ accounts for load rejection.

4. Realizability and the Low-Pass Correction

Expanding $G_1(s)$:

$$G_1(s) = \frac{J_{\text{eff}}L}{K_t}\,s^2 + \frac{J_{\text{eff}}R + BL}{K_t}\,s + \frac{BR + K_tK_b}{K_t} \tag{8}$$

This is a *polynomial* in s (improper transfer function): its numerator degree (2) exceeds its denominator degree (0). Physically, applying $G_1(s)$ requires twice-differentiating the reference ω_{ref}, which amplifies noise and is non-causal in practice.

To make the feedforward realizable, a second-order low-pass filter $(\tau s + 1)^{-2}$ is appended:

$$G_{ff}(s) = \frac{G_1(s)}{(\tau s + 1)^2} \tag{9}$$

The filter introduces two poles and shifts relative degree to zero, making $G_{ff}(s)$ proper. However, choosing τ requires careful tuning and the resulting filter still demands precise knowledge of all motor parameters (R, L, K_t, K_b, J, B), making it fragile and impractical outside of MATLAB/Simulink environments.

5. Polynomial Regression as a Practical Alternative

Motivation

Rather than implementing the model-based inverse filter, we adopt a *data-driven* approach: measure the steady-state relationship between voltage and speed experimentally, invert it, and fit a polynomial.

Data collection

The motor is driven at each voltage level $V \in \{-12, -9, \ldots, 9, 12\}$ V. The simulation waits for steady state (rolling 1 s window with $\sigma_\omega < 0.5$ rad/s and drift < 0.3 rad/s), then records $N = 100$ noisy speed samples per level, yielding a dataset $\{(V_k, \omega_k)\}$.

Inversion and zero-bias fit

Because the motor is symmetric and exhibits no offset at $\omega = 0$, the inverse map $V = f(\omega)$ is constrained to pass through the origin. A zero-intercept polynomial of degree n is fitted by solving:

$$\min_{\mathbf{c}} \left\| X\mathbf{c} - \bar{V} \right\|^2, \quad X_{ij} = \bar{\omega}_i^j, \;\; j = 1, \ldots, n \tag{10}$$

using `numpy.linalg.lstsq`, where $\bar{\omega}_i$ and $\bar{V}_i$ are the per-level means. With $n = 2$ the result is:

$$\hat{V}(\omega) = c_1\,\omega + c_2\,\omega^2 \tag{11}$$

The fitted coefficients are $c_1 \approx 0.3588$ V s/rad and $c_2 \approx 0$, consistent with the theoretical steady-state gain $\frac{RB}{K_t} + K_b = 0.35$ V s/rad. At runtime the feedforward voltage is simply:

$$V_{ff}^{\text{poly}}(\omega_{\text{ref}}) = \hat{V}(\text{clip}(\omega_{\text{ref}}, \omega_{\min}, \omega_{\max})) \tag{12}$$

This requires *no parameter knowledge*, is trivially fast to evaluate, and generalises to any actuator with a monotone steady-state characteristic.

6. Experimental Comparison

Eight closed-loop experiments compare controllers on two reference profiles: a constant step ($\omega_{\text{ref}} = 10$ rad/s) and a quintic trajectory that smoothly ramps from 0 to 10 rad/s between $t = 0.5$ s and $t = 3.5$ s with zero boundary velocity and acceleration. All controllers share $K_p = 5$, $K_i = 2$.

Table 2 reports the RMS tracking error over 5 s. The best result is highlighted in green.

Discussion

Pure PI achieves basic regulation but carries a steady-state offset under gravity load. Adding the analytical gravity feedforward (Section 2) reduces the error marginally on constant references but still leaves the integral to compensate model mismatch.

Table 2: RMS tracking error (rad/s) for each controller and reference.

Controller	Const. ref	Trajectory
PI only	0.495	0.322
PI + gravity FFW	0.450	0.279
PI + poly FFW	0.376	0.171
PI + poly FFW + gravity FFW	0.322	**0.033**

The polynomial feedforward provides a larger benefit: by pre-computing the steady-state voltage for any ω_{ref}, it removes the dominant tracking lag and allows the PI to act only on residual disturbances.

Combining both feedforwards yields the best performance. On the quintic trajectory the RMSE drops to **0.033 rad/s** — an order of magnitude below pure PI — because the polynomial term handles the voltage demand and the gravity term compensates the angle-dependent disturbance, leaving the integrator nearly idle. The constant-reference experiments show smaller absolute differences because the integrator eventually winds up to eliminate any steady bias, narrowing the gap between strategies.

7. Conclusion

A complete simulation pipeline was developed for a BDC motor driving a vertical rod. The model-based inverse feedforward, while theoretically exact, is non-realizable without parameter knowledge and additional filtering. Polynomial regression on steady-state input–output data provides a practical, model-free substitute that, combined with an analytical gravity term, reduces trajectory-tracking error by $\sim$10× compared to PI alone. The pipeline — data collection, curve fitting, controller design, and benchmarking — is fully automated and reproducible from the project root.